DL3004 DWBI Formal Requirements & Specification

Project:	U.S. Bulk Power Generation, Smokestack Air Emissions, and Synoptic Weather Warehouse	**Course:**	DL3004 Data Warehousing & Business Intelligence
Institution:	FAST National University of Computer & Emerging Sciences	**Phase:**	Phase 1: Design Specification (30% Weight)
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1. Executive Summary & Problem Statement

Electricity generation in the United States is measured by the **Energy Information Administration (EIA)**, continuous smokestack emissions by the **Environmental Protection Agency (EPA)**, and ambient meteorology by the **National Oceanic and Atmospheric Administration (NOAA)**. These federal agencies publish data in disparate formats (Excel workbooks, fixed-width text files, CSV, and REST APIs) and at differing temporal grains (hourly stack unit measurements, monthly facility aggregates, and hourly grid balancing telemetry).

Without an enterprise dimensional data warehouse, environmental and energy analysts cannot reliably answer fundamental questions such as: which power plants produce the highest carbon intensity per megawatt-hour produced, how extreme temperature swings impact boiler heat rates, and which regional balancing authorities achieve the highest clean renewable penetration. This project designs, implements, and analyzes an end-to-end PostgreSQL data warehouse resolving these operational challenges.

2. Core Analytical Business Questions

#	Analytical Question	Underlying Dimensional Integration	Target BI Visual & Metric
BQ1	Which generating facilities exhibit the highest CO_2, SO_2, and NO_x emissions intensity per gross MW generated?	Join `dim_plant` to `fact_hourly_emissions_weather`; calculate emissions per MWh.	Ranked bar chart & carbon intensity KPI gauge (tons CO_2/MW).
BQ2	How do extreme ambient temperatures (heatwaves & winter freezes) impact fossil generator heat rates and operational capacity?	Spatial join between `dim_plant` and nearest `dim_weather_station` (≤50 km) linked to hourly load.	Scatter plot of Heat Rate (Btu/kWh) vs. Surface Temp (°C) with linear regression fit.
BQ3	Which regional balancing authorities have the highest renewable energy penetration	Aggregate `fact_regional_grid_demand` joined with	100% stacked area chart and geographic choropleth map.

#	Analytical Question	Underlying Dimensional Integration	Target BI Visual & Metric
	(wind & solar) and lowest net interchange dependency?	`dim_balancing_authority` over time.	
BQ4	How do power plant operational states evolve across generator asset lifecycles, and which units are retiring?	**SCD Type 2** tracking in `dim_generator` via `operating_status` and effective dates.	Timeline waterfall visual and historical capacity transition matrix.
BQ5	How do generation dispatch and station service load respond to regional demand peaks?	Cross-fact analysis between `fact_hourly_emissions_weather` and `fact_regional_grid_demand`.	Dual-axis line and bar chart comparing real-time dispatch against forecast peak.

3. Multi-Source Ingestion Inventory

Source System	Agency	Extraction Format	Natural Grain	Target Ingestion Layer
EIA-860	U.S. EIA	Excel Workbook (.xlsx)	Annual facility & generator asset inventory	`project_staging.raw_eia860_generator`
EIA-860 (6.1)	U.S. EIA	Excel Workbook (.xlsx)	Boiler-to-generator environmental associations	`project_staging.raw_eia860_enviro_assoc`
EIA-923	U.S. EIA	Excel Workbook (.xlsx)	Monthly plant fuel consumption & net generation	`project_staging.raw_eia923_monthly_generation`
EPA CAMPD	U.S. EPA	Bulk CSV / REST API	Hourly continuous emission monitoring (CEMS) per stack	`project_staging.raw_epa_hourly_emissions`
NOAA ISD	NOAA NCEI	Fixed-Width Text / CSV	Hourly synoptic surface weather observations	`project_staging.raw_noaa_isd_weather`
EIA-930	U.S. EIA	JSON REST API / CSV	Hourly regional grid demand and fuel mix telemetry	`project_staging.raw_eia930_hourly_balance`

4. Complete Medallion Pipeline Architecture

The project adheres to an 8-stage Medallion architecture orchestrated entirely via Python and PostgreSQL:

1. **Data Sources:** Extraction of public microdata across federal agencies.
2. **Pipeline Orchestrator:** Modular Python script managing dependencies, unique `batch_id` UUID generation, and pipeline logs.
3. **Bronze Layer (`project_staging`):** Schema preserving 100% of raw input data as `VARCHAR` to guarantee zero ingestion failures.
4. **Data Quality & Dead-Letter Quarantine (`quarantine`):** Automated SQL validation engine catching string sentinels, thermodynamic bounds violations, and referential errors, routing corrupt rows to `quarantine.rejected_records` with raw JSON payload.
5. **Silver Layer:** Data cleaning, unit standardization (MWh, MMBtu, lbs, short tons, °C), geospatial Haversine station joins, and SCD Type 2 dimension processing.
6. **Dimensional Data Warehouse (`project_dw`):** Kimball star schema consisting of 5 conformed dimensions and 3 fact tables at distinct analytical grains.
7. **Gold Layer & Reporting Mart (`project_reporting`):** Materialized and analytical views optimized for Power BI.
8. **Analytics & BI:** Multi-dimensional OLAP queries (`ROLLUP`, `CUBE`), `EXPLAIN ANALYZE` query indexing, and interactive Power BI executive dashboards.

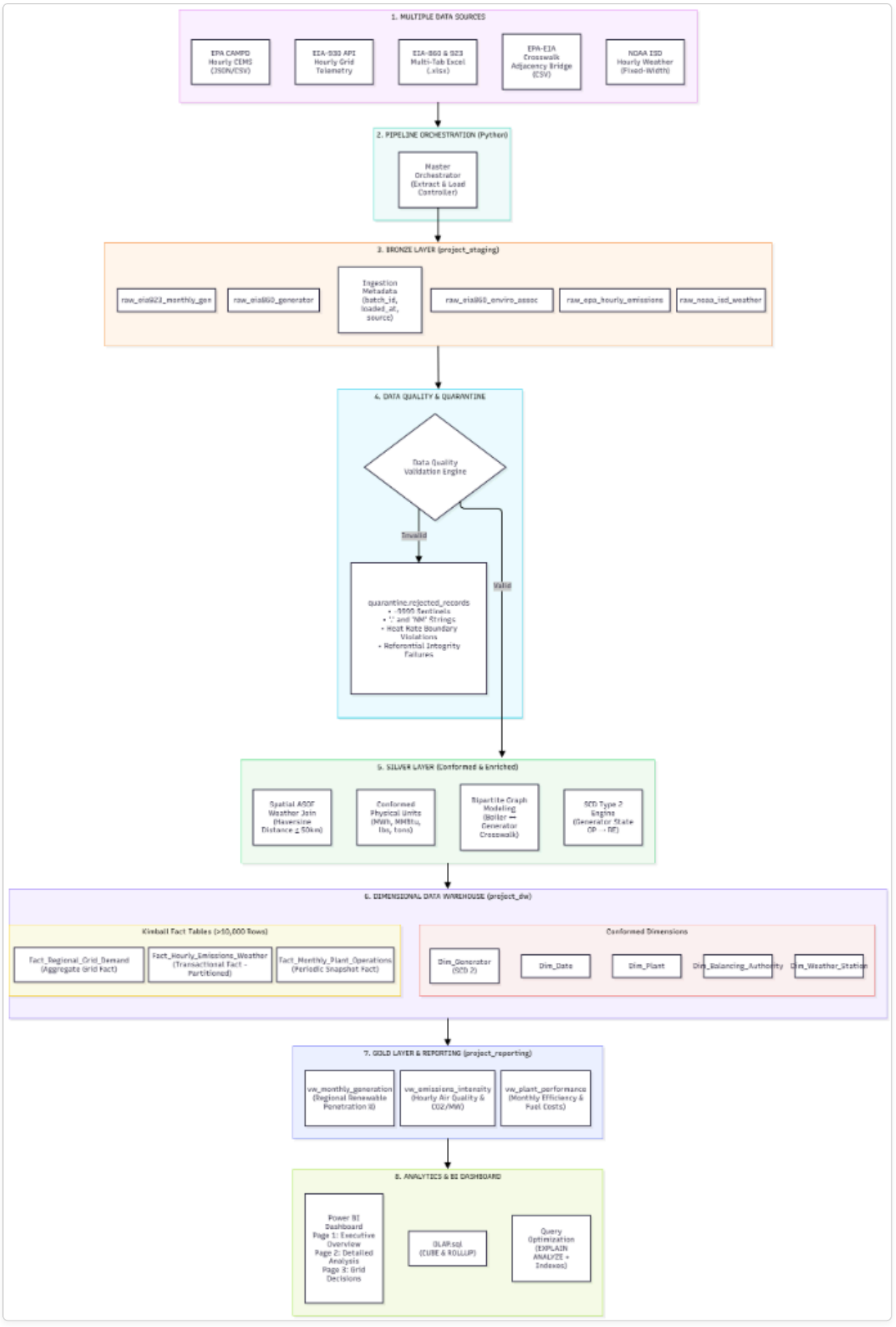


Figure 1: End-to-End Medallion Pipeline Architecture and Schema Flow

5. Kimball Dimensional Model (Star Schema)

The enterprise warehouse is structured around the Kimball dimensional bus matrix:

- **Conformed Dimensions (5):** `dim_date` (smart integer PK `date_key` YYYYMMDD), `dim_plant` (natural key `orispl_code`), `dim_generator` (**SCD Type 2** tracking `operating_status` with `row_effective_date` and `is_current`), `dim_balancing_authority` (natural key `ba_code`), and `dim_weather_station` (natural key `station_id`).
- **Fact Tables (3):**
  - `fact_hourly_emissions_weather`: Transactional fact partitioned by `date_key` at unit-hour grain.
  - `fact_monthly_plant_operations`: Periodic snapshot fact tracking net generation, thermal heat rates, and fuel stocks.
  - `fact_regional_grid_demand`: Aggregate grid fact tracking hourly regional load and fuel mix.

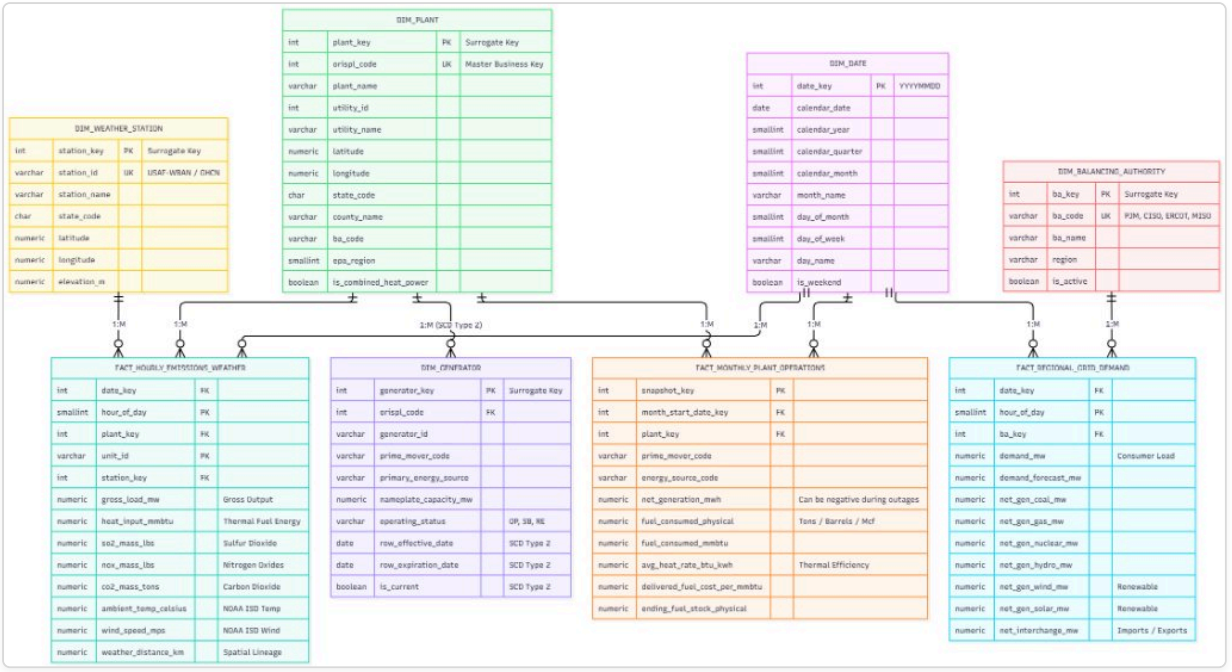


Figure 2: Physical Kimball Star Schema Entity-Relationship Blueprint

6. Deliverables & Rubric Compliance Matrix

Phase & Weight	Deliverable Artifact	Location in Repository	Status
Phase 1: Design (30% Marks)	Requirements Specification (PDF)	`01_Project_Documentation/Requirements.pdf`	COMPLETE
	Star Schema Diagram (PDF/Image)	`01_Project_Documentation/Schema.pdf`	COMPLETE
	Multi-Tab Master Data Dictionary (Excel)	`01_Project_Documentation/Data_Dictionary.xlsx`	COMPLETE
	Comprehensive Proposal & Design Pack	`Phase_1_Submission/`	COMPLETE
Phase 2: Implementation (40% Marks)	PostgreSQL DDL & Schemas	`04_Data_Warehouse/warehouse.sql`	DESIGN READY
	ETL Ingestion & Quarantine Pipeline	`03_ETL/staging.sql`, `quality_checks.sql`	DESIGN READY
	Data Cleansing & Staging Data	`02_Data/raw_data/`, `cleaned_data/`	DESIGN READY
Phase 3: Analytics & BI (30% Marks)	OLAP Queries (ROLLUP, CUBE)	`05_Analytics/OLAP.sql`	DESIGN READY
	Reporting Views	`05_Analytics/Views.sql`	DESIGN READY
	3-Page Decision Power BI Dashboard	`06_Dashboard/PowerBI.pbix`	DESIGN READY

7. Verification Checklist Before Submission

- ☑ **Repository Structure:** Strictly conforms to course guidelines (`01_Project_Documentation`, `02_Data`, `03_ETL`, `04_Data_Warehouse`, `05_Analytics`, `06_Dashboard`).
- ☑ **Data Dictionary:** Authoritative 7-tab Excel file with metadata, staging, dimensions, facts, quarantine, views, and lineage/codes.
- ☑ **Diagrams:** High-resolution vector/raster diagrams embedded in all submission files.
- ☑ **Zero Paid Infrastructure:** All components run on local PostgreSQL 16+, Python 3.10+, and Power BI Desktop.